

[DC] Privacy in VR: Empowering Users with Emotional Privacy from Verbal and Non-verbal Behavior of Their Avatars.

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ABSTRACT

The next paradigm for social and collaborative VR applications is heavily based on the real time behavior of tracked avatars for rich communication. Industry giants are actively focusing on producing photo-realistic avatars with facial and body movement tracking of real time captured verbal and non-verbal cues. This enables applications to convey the user's actual emotive expressions and behaviors through a virtual avatar, enabling the exchange of rich communication. However, the capability exposes individualized patterns of users without providing any opportunity to filter out undesired tracking before expressing them in front of a crowd, and thus can expose privacy problems. In order to address the trade-off between rich communication and privacy in VR, my PhD dissertation is focusing on providing rich user behaviors using verbal and non-verbal cues through VR avatars, while also providing a protective layer to eliminate unwanted or unneeded leaking of emotions or movements, and thus providing a more-trusted collaborative VR experience.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Virtual Reality—Behavior tracked avatar embodimentFacial emotion privacy;

1 INTRODUCTION

Virtual reality (VR) is regarded as a promising tool for remote collaboration and communication, due to the evolution of VR technology, such as high-resolution rendering and tracking quality, and with multi-sensory feedback. Now, VR is even able to deliver fine emotional cues with tracked non-verbal behaviors via facial expressions and body-gestures for rich communication. In computer-generated environments, VR users are encouraged to disclose possibly sensitive information about themselves by providing both implicit (emotion or behavior pattern) or explicit (name or ID) identity cues to support trusted social networks and eliminate potential malicious usage. In this research, our scope covers only the implicit exposure. In short, implicit self-disclosure, such as emotions in VR, can establish an alliance with others, create and maintain relationships, make self-expression and self-clarification, or achieve social validation as we do in the real world [2–4]. In line with this perspective, human emotions play a vital role in communication in the real world, and researchers assume this will also be true in VR [5, 9]. Hence, developers are capturing every possible motion and emotional cue to increase the fidelity level of the application [10], since emotional cues can be categorized as written or spoken language, facial display, posture and other physiological activities [9]. Among the above-mentioned cues, facial expressions are highly correlated with the actual psychological status of the user, and can be displayed consciously or unconsciously by the user. These are evidently indicated through eye and mouth movements [5].

However, even though these photo-realistic embodied avatars enable a rich communication between the collaborators, it can in-

roduce disadvantageous and inconvenient situations to the avatar user depending on the context. For example, in a situation like a job interview, interviewers might easily identify negative traits like nervousness which could badly affect the final decision.

Unfortunately, most VR users are still not aware of the possible violation of privacy that could happen through these kinds of applications [1]. There are some studies about the overall legal background related to VR, stating that everyone has the right to privacy, and this disclosure is one aspect of privacy that could possibly be violated by above-mentioned applications. They also suggest that the technology is progressing at an exponential rate, while the slow-moving legal system fails to catch up [6, 10]. Although the legal system is not focusing on *emotional* privacy at the moment, the technologists have a responsibility to eliminate, or at least minimize, any potential threats to user privacy.

Hence, this dissertation is focusing on enabling an additional layer between emotional-cue tracking and avatar embodiment systems to provide the ability to avoid getting tracked outside their explicit desired level, while at the same time ensuring rich interactivity for collaborative VR environments. To my knowledge, the related research has not yet been sufficiently conducted.

The rest of this paper will present the intended objectives and goals of the research, and the current status.

2 RESEARCH OBJECTIVES AND GOALS

The objective of this research is to determine an effective way to filter the tracked user's verbal and non-verbal behaviors, such as facial expressions, physiological signals, and body gestures used to drive their avatar, with a privacy layer, masking the threats to the discovery of the individual while still providing the expressiveness needed for effective collaboration and communication. Research questions aim to address with this research are:

- How can we ensure a user's emotional privacy, when expressing emotions through avatars by tracking real-time verbal and non-verbal behavior of users?
- How can we identify undesired emotions that need to be filtered according to the context in real time?
- What is the appropriate way to replace undesired emotions in real time without harming the collaborative environment?

I am still in the early stages of my studies (proposal preparation), and currently plan to execute in three basic stages. In order to conduct the research, a scenario of a professional medical counselling session is proposed. The target user of the emotion controlling system will be a medical counselor.

2.1 Identify Undesired Emotions According to Context

Undesired emotions based on the above-mentioned counselling scenario will be identified according to a multi-model structure based on a Machine Learning approach. Emotional and behavioral cues from the target (counselor), observer (patient) and the environment will be analyzed. Under these three main subjects, facial emotional cues, body gestures and posture, voice prosody and environmental noise or anomalies will be used to identify undesired emotions of the counselor.

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2.2 Replace Undesired Emotions without Interrupting Collaboration

Replacing undesired emotions is a real-time task, and in order to replace the filtered-out emotion with a proper emotion that is suitable to the situation, understanding the conversation and collaboration context is important. Initially, reinforcement-learning-based mechanisms will be used. Studies will be carried out to find solutions for controlling the virtual avatar based on three options: 1. The user selects the emotion for the avatar, 2. A reinforcement learning based algorithm selects a suitable emotion, and 3. A hybrid solution of the algorithm suggesting several emotions, and user confirming one. This will give user the ability to stay away from perfectly tracked movements without degrading collaborative interaction. Performing this task without interrupting the real time interaction will be a challenge in this task.

2.3 Introduce a Framework to Increase the User Awareness about the Tracked Features.

The intention of this framework is to present the user with all the tracked and analyzed information, such as the current user emotion and any detected anomalies in overall emotion. ex:if user is getting angry suddenly in between a calm collaboration this could be an anomaly which possibly harm the nature of the collaboration. Hence system could warn user to stay away from tracking if desired. This will help user to decide and define when to stay away from getting tracked. It is identified that make the user aware of the actual situation help them to get precautions to protect their privacy. [3]

3 CURRENT STATUS

As the current design status, a facial emotion recognition model based on a Convolutional Neural Network (CNN) has been designed, trained and tested against publicly available emotion recognition datasets. The CNN model classifies facial emotions into eight basic emotions: 1. Neutral, 2. Calm, 3. Happy, 4. Sad, 5. Angry, 6. Fearful, 7. Disgusted, 8. Surprised. The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) has been used to train the CNN model. [7] Currently a 75% accuracy has been achieved for the constrained facial emotion recognition, and a study is planned to test the performance of the model in real-world scenarios, and to check the accuracy of the emotion recognition model against human responses. This analysis is important as human facial emotion is a complex expression, blending several different emotions. Figure 1 shows the analyzed results of the Harry Potter movie sorting hat scene, and Figure 2 shows the Louise virtual model emotions using the emotion recognition CNN model [8, 11]. We are planning to conduct a study with a photo-realistic-facial-expression-enabled avatar to analyze human and machine learning based model's ability to identify emotions from a virtual avatar.

4 CONCLUSION

This research, which is in the early stages, is proposed and planned to analyze the ability to empower tracked avatar users with emotional privacy. It is designed to give the user the capability of avoiding explicit constant facial-movement tracking without interrupting or disturbing the collaborative nature of VR applications. I would appreciate feedback of the system and possible ways to handover more control and /or power to the user without harming real-time collaboration.

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Figure 1: Results of the CNN model for facial emotion recognition of the sorting hat scene in Harry Potter movie.

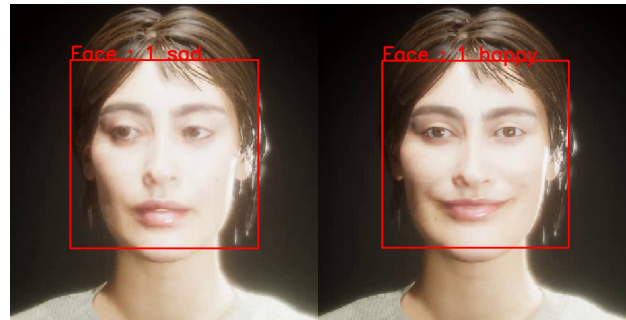


Figure 2: Results of the CNN model for facial emotion recognition of photo-realistic virtual avatar model Louise.

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